

Summary And Recommendations

This exploratory data analysis (EDA) provides insights into customer behavior and the key factors driving churn within the Telco Customer dataset. Through a series of visualizations—including churn distribution, service-level comparisons, tenure trends, payment patterns, and contract types—the analysis highlights clear behavioral patterns associated with customer retention and churn.

Overall churn is noticeably higher among customers using **month-to-month contracts**, **electronic check payments**, and those lacking **Internet-related security services** such as OnlineSecurity, OnlineBackup, and TechSupport. Tenure-based charts show that long-term customers are significantly less likely to churn, indicating the importance of early engagement and retention interventions. Additionally, stacked bar plots across multiple service categories reveal that customers who actively use additional services generally show higher churn, especially when those services are less supportive or have limited value.

These insights collectively indicate that churn is strongly influenced by contract flexibility, billing method, tenure, and availability of support or security-related services. The analysis helps identify priority areas for improving customer experience, strengthening service offerings, and designing targeted retention strategies.

Overall Churn Landscape

The dataset shows that approximately **26%** of customers have churned, while **74%** continue with the service. This imbalance highlights that although the majority remain loyal, over a quarter of customers still represent a significant revenue leakage. The churn variable was further analyzed across demographic, service, and payment-related fields to uncover deeper drivers.

Tenure & Churn Duration Patterns

Tenure-based visualizations show a strong negative correlation with churn:

- Customers with a tenure of **0–12 months** contribute to **nearly 50% of total churn**.
- Long-tenure customers (5–10 years) show a churn rate of **less than 10%**.

This suggests customer loyalty increases significantly with time, highlighting the importance of **early-stage retention programs**.

Contract Type & Churn

Contract type shows one of the strongest predictors:

- **Month-to-month** customers have a churn rate of **43%**.
- **One-year** contract customers churn at only **11%**.
- **Two-year** contract customers churn at just **3%**.

Charts clearly show that short-term customers are much more unstable and sensitive to dissatisfaction.

Payment Method Influence

Analysis of the payment method reveals:

- Customers paying via **Electronic Check** show the highest churn at **45–48%**.
- Those using **Bank transfer or Credit card** exhibit churn rates below **15%**.

This suggests that electronic check users may experience more issues or face higher monthly charges, prompting disengagement.

Service Subscriptions & Churn Patterns

The stacked plots for multiple services (PhoneService, InternetService, TechSupport, StreamingTV, etc.) reveal that churn is heavily concentrated among customers subscribed to internet-related and add-on services:

- Customers without **OnlineSecurity** show a churn rate of **~40%**, compared to **15%** for those who have it.
- Customers without **TechSupport** churn at **~35%**, whereas those with support churn at **~17%**.
- Internet-only customers (no additional services) show higher instability than bundled-service users.

These patterns suggest that **support and security features significantly improve customer stickiness**.

Gender, Senior Citizen & Demographics

Although demographic variables show mild effects:

- Senior citizens churn at **~42%** compared to **~24%** among non-senior users.
- Gender, however, shows no meaningful difference, with churn around **26%** for both males and females.

Key Strategic Insights

The overall analysis demonstrates that churn is driven primarily by:

1. **Short-term contracts**
2. **Electronic check billing**
3. **Low tenure**
4. **Lack of technical support or security add-ons**
5. **Higher monthly charges (shown in distribution charts)**

In contrast, customers with long-term contracts, bundled support services, and predictable billing methods show significantly higher retention.